

TSD 2026 — Software Testing Project

Team Crousty

Application: Demo Web Shop · <http://localhost:3001>

Submission: 9 July 2026

1. Application Overview

Demo Web Shop — a single-page e-commerce application

Detail	Value
Stack	Node.js · json-server 0.17.4 · Vanilla JS
Routing	Hash-based SPA (<code>/#/login</code> , <code>/#/cart</code> , ...)
URL	<code>http://localhost:3001</code> (local)

Main features tested

- **User Authentication** — login / error handling
- **Product Search** — keyword search, results display
- **Shopping Cart** — add to cart, quantity, notifications
- **Checkout Process** — guest and registered flows
- **Boundary validation** — qty limits, password length

2. Test Strategy

10 Manual Test Cases (TC-001 → TC-010)
↓ ↓ ↓
Selenium (3) Robot FW (2) Postman (API)
TC-001/002/003 TC-001/004 REST endpoints

Tool	Language	TC covered
JUnit 5	Java	Rating unit tests (Lab 2)
Selenium WebDriver 4.18.1	Java	TC-001, TC-002, TC-003
Robot Framework 7.1.1	Python	TC-001, TC-004
Postman	—	REST API layer

Automation criteria: stable UI flow · clear assertion · high regression value

3. Manual Test Cases

TC ID	Title	Type	Status
TC-001	Successful login	Positive	✓ PASS
TC-002	Search existing product	Positive	✓ PASS
TC-003	Add item to cart	Positive	✓ PASS
TC-004	Login with wrong password	Negative	✓ PASS
TC-005	Search non-existing product	Negative	✓ PASS
TC-006	Checkout with empty cart	Negative	✓ PASS
TC-007	Password minimum length	Boundary	✓ PASS
TC-008	Add zero/negative quantity	Boundary	✗ BUG-001
TC-009	Full checkout – Guest	Flow	✓ PASS
TC-010	Full checkout – Registered	Flow	✓ PASS

9/10 PASS · 1 defective (BUG-001)

4. Defect Report — BUG-001

Title: System accepts quantity 0 silently — no validation error

Field	Value
Severity	Medium
Component	Shopping Cart
Steps	Open product page → enter qty 0 → click Add to cart
Expected	Error: "Quantity must be greater than 0"
Actual	✔ Success toast shown, but cart count stays at 0
Status	Open

Silent success with incorrect data is **harder to detect** than a crash and more confusing for end users — the cart appears to have accepted the item.

5. Selenium WebDriver Automation

Tool: Selenium 4.18.1 + JUnit 5 · Java

Test	TC	Result
TC001_shouldLoginWithValidCredentials	TC-001	✓ PASS
TC002_shouldDisplayResultsForLaptop	TC-002	✓ PASS
TC003_shouldAddProductToCart	TC-003	✓ PASS

```
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 15.63 s -- in DemoWebShopTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 37.205 s
[INFO] Finished at: 2026-07-06T13:39:00+02:00
[INFO] -----
```

6. Robot Framework Automation

Tool: Robot Framework 7.1.1 + SeleniumLibrary 6.7.1 · Python

Test	TC	Type	Result
TC-001 Valid Login Shows User Email In Header	TC-001	Positive	PASS
TC-004 Invalid Login Shows Error Message	TC-004	Negative	PASS

Keywords: Open Demo Web Shop · Navigate To Login Page · Fill Login Form · Verify User Is Logged In · Verify Login Error Message

Test Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	2	2	0	0	00:00:07	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
login	2	2	0	0	00:00:07	
negative	1	1	0	0	00:00:04	
positive	1	1	0	0	00:00:03	
TC-001	1	1	0	0	00:00:03	
TC-004	1	1	0	0	00:00:04	

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Demo Shop Tests	2	2	0	0	00:00:07	

Test Details

AllTagsSuitesSearch

Status: 2 tests total, 2 passed, 0 failed, 0 skipped

Total Time: 00:00:06.911

Name	Documentation	Tags	Status	Message	Elapsed	Start / End
Demo Shop Tests: TC-001 Valid Login Shows User Email In Header	Positive test: a registered user logs in with correct credentials and sees their email address in the page header.	login, positive, TC-001	PASS		00:00:02.636	20260707 13:46:14.610 20260707 13:46:17.246
Demo Shop Tests: TC-004 Invalid Login Shows Error Message	Negative test: a user who provides a wrong password sees an error message and is not redirected to the home page.	login, negative, TC-004	PASS		00:00:04.275	20260707 13:46:17.248 20260707 13:46:21.523

7. Postman API Tests

Endpoints under test (`http://localhost:3001`):

Endpoint	Method	Purpose
<code>/products</code>	GET	List all products
<code>/products?name_like=Laptop</code>	GET	Keyword search
<code>/cart</code>	GET / POST	Read / add items
<code>/users</code>	GET	List registered users

Results to be added after Lab 6 (2026-07-09).

Approach: Postman Collection with environment variables (`{{BASE_URL}}`) and automated test scripts validating status codes, response structure, and data integrity.

8. Results Summary

Lab	Tool	Tests	✓ Pass	✗ Fail
Lab 3	Manual (Demo Web Shop)	10	9	1
Lab 4	Selenium WebDriver	3	3	0
Lab 5	Robot Framework	2	2	0
Lab 6	Postman	TBD	—	—

Lab 2 (JUnit — Rating class) is a standalone exercise, not a Demo Web Shop test.

Total automated: 5 tests · 5 passed · 0 failed · 1 defect open (BUG-001)

Coverage RatingTest				
Element ^	Class, %	Method, %	Line, %	Branch, %
sk.fri	50% (1/2)	50% (1/2)	72% (8/11)	81% (13/16)
Main	0% (0/1)	0% (0/1)	0% (0/3)	0% (0/2)
Rating	100% (1/1)	100% (1/1)	100% (8/8)	92% (13/14)

9. Lessons Learned

1. **Explicit waits are mandatory for SPAs** — async fetch calls break any fixed sleep
2. **Test isolation requires intent** — `localStorage.clear()` prevented TC-001 ↔ TC-003 state leak
3. **Robot Framework is more readable** — keyword syntax bridges dev/non-dev gap; Selenium gives more control for complex scenarios
4. **Maven Wrapper (`mvnw.cmd`) enables portability** — zero-config Maven for any new machine
5. **Cross-OS node_modules are incompatible** — WSL vs Windows binary mismatch; always install in the target OS
6. **Silent bugs are worse than crashes** — BUG-001 looked like success; only boundary testing revealed it

Conclusion

The Demo Web Shop project exercised the **full testing lifecycle**:

manual design → unit testing → UI automation → API testing

9/10 manual tests pass · 1 confirmed defect · 11 automated tests green

Each tool has its place: JUnit for logic, Selenium for Java-team automation, Robot Framework for readable acceptance tests, Postman for API contracts.

